Pick a real nozzle case using either a LOX/LH2 approximation w/ $\gamma \approx 1.2$, or use cold nitrogen $\gamma = 1.4$ for simplicity.

Given: $P_o = 500\,psi$, $\frac{A_e}{A^*} = 4.0$

Without using CFD, calculate:

1. Throat Conditions 2. Exit Mach # 3. Exit Pressure, Temp., Velocity

Assumptions: Steady, isentropic, ideal-gas flow through CD nozzle

$\gamma = 1.4$, $R = .297 \frac{kJ}{kg\,K}$, $T_o = 300K$

Solution: ① $M^* = 1 \longrightarrow \frac{T}{T_o} = \frac{1}{1+\frac{\gamma-1}{2}M^2} \rightarrow T = \left[\frac{1}{1+\frac{\gamma-1}{2}M^2}\right] T_o$

$$T = \left[\frac{1}{1+\frac{1.4-1}{2}(1)^2}\right] 300 = 250K \rightarrow \boxed{T^* = 250K}$$

$$\frac{P}{P_o} = \left[1+\frac{\gamma-1}{2}M^2\right]^{-\frac{\gamma}{\gamma-1}} \longrightarrow P = P_o\left[1+\frac{\gamma-1}{2}M^2\right]^{-\frac{\gamma}{\gamma-1}} = 500\left[1+\frac{1.4-1}{2}(1)^2\right]^{-\frac{1.4}{1.4-1}}$$

$\boxed{P^* = 264\,psi}$ $= 1820\,kPa$

$$\rho^* = \frac{P^*}{RT^*} = \frac{1820}{.297\,(250)} = \boxed{24.5\ \frac{kg}{m^3}}$$

② $$\frac{A_e}{A^*} = 4.0 \rightarrow \frac{A_e}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

$$4.0 = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

$$4.0 = \frac{1}{M}\left[\frac{2}{1.4+1}\left(1+\frac{1.4-1}{2}M^2\right)\right]^{\frac{1.4+1}{2(1.4-1)}}$$

$$4.0 = \frac{1}{M}\left[0.833\left(1+0.2M^2\right)\right]^3$$

$$\boxed{M_e = 2.94}$$

We will pick the supersonic branch since $M_e > 1$

③ Using known isentropic relations, $M_e = 2.94$

$\underline{P_e}$: $\dfrac{P_e}{P_0} = \left[1 + \dfrac{\gamma - 1}{2} M^2\right]^{-\frac{\gamma}{\gamma-1}} \rightarrow P_e = 500\left[1 + \dfrac{1.4-1}{2}(2.94^2)\right]^{-\frac{1.4}{1.4-1}}$

$\boxed{P_e = 14.9 \text{ psi}}$

This makes sense!
If we had sustained supersonic expansion within the diverging region of the nozzle, our high M_e explains our low P_e in accordance w/ the cons. of energy

$\underline{T_e}$: $\dfrac{T_e}{T_0} = \dfrac{1}{1 + \frac{\gamma-1}{2} M_e^2} \rightarrow T_e = T_0\left[\dfrac{1}{1 + \frac{\gamma-1}{2} M_e^2}\right]$

$T_e = 300\left[\dfrac{1}{1 + \frac{1.4-1}{2} 2.94^2}\right] = \boxed{110 \text{ K}}$

Also makes sense per the conservation of energy!

$\underline{V_e}$: $V_e = M_e\sqrt{\gamma R T_e} = 2.94\sqrt{1.4(297)(110)} = \boxed{629\ m/s}$

Makes sense : Velocity should be high since our Mach # is high per

$$M = \frac{V}{a}$$